Who Still Hears the Warning?
Persistent Dissent in a Networked Evacuation Game

Initial proposal A — November 8, 2026

Abstract

How does repeated disagreement affect who agents listen to and whether they take the correct action? We propose an agent adaptation of an established human evacuation game to investigate changes in consultation, message uptake, and collective outcomes without assuming that repetition helps or harms. Live agents exchange local messages and independently decide whether to incur an evacuation cost or remain exposed to a possible disaster. A controlled warning source receives a truthful observation; its communication schedule varies while other agents interact freely. Fixed and agent-selected consultation conditions distinguish failures of uptake from failures of delivery. Incorrect-warning and disaster-free controls prevent treating persistence or evacuation as inherently beneficial. We will measure consultation choices, warning delivery, and state-appropriate actions separately. This proposal specifies a pilot design rather than results; its potential contribution is a test of the consequences of endogenous communication access in an action-based network task.

1 Introduction

The research question is: *How does repeated disagreement affect who agents listen to and whether they take the correct action?* In the evacuation game, this becomes a test of whether repeating a warning changes consultation and state-appropriate evacuation. Repetition may improve uptake, have no effect, or reduce access; these are competing possibilities. An ignored warning is not by itself exclusion. We require observable choices that prevent later delivery, and retain separate measures of hearing, relaying, and acting. This connects collective safety to the network's communication decisions rather than to a single model responding to scripted peer opinions.

2 Related work

The evacuation experiment provides a precedent for studying communication and costly safety actions together [1]. Schachter's study of persistent disagreement motivates examining communication and rejection, but its dissenter was not an objectively correct safety expert [2]. Contemporary LLM deliberation already studies correct minorities and participation [3]. Our candidate question concerns how chosen communication access changes later safety decisions; neither persistence nor minority recovery alone is a novelty claim.

3 Proposed methods

3.1 Established task and proposed adaptation

Shirado, Crawford and Christakis studied a networked evacuation game: participants exchanged local Safe/Danger signals and chose whether to evacuate. Evacuation was costly; remaining during a disaster lost the endowment. One informant knew the true state. Network size and topology were manipulated [1]. We adapt this task rather than inventing a scenario; we do not predict that models reproduce human psychology.

The proposed simulator has ten live model agents on a fixed degree-four ring, separate histories, and local messages. A balanced hidden state determines disaster versus no disaster. One randomly placed informant receives an authenticated state observation; others know only that an informant exists. All agents share the payoff rules.

Own payoffs are 1 point for evacuation, 0 for remaining during disaster, and 2 for remaining without disaster. As in the human task, surviving agents also receive 0.1 point per other agent taking the state-appropriate action. Points are evaluation signals, not evidence of human-like motivation.

For reproducibility, replace real-time play with synchronous rounds and a simulator-selected, undisclosed terminal round. Freeze its distribution and maximum after calibration. Every active agent chooses a signal and stay/evacuate; actions and messages are applied together after all responses. Evacuation is irreversible. Exited agents cannot transmit, while previously delivered messages remain in recipients' histories. Use structured outputs and a fixed message budget. These scheduling and model-interface choices are adaptations, not a direct replication.

3.2 Research scope: fixed settings and experimental variables

The overarching question is: *How do a message's source, repetition, and supporting evidence affect its spread and influence on collective decisions?* The evacuation game is the concrete test environment, not the boundary of the research question. Evidence access is controlled in the initial experiments; any later manipulation of evidence quality requires a separately specified contrast.

Factor	Status and proposed conditions
Fixed environment	Ten agents, degree-four ring, payoff rules, separate histories, synchronous delivery, irreversible evacuation.
Fixed within comparison	Message budget, task instructions, evidence access, actual permissions, sampling settings, and timing distribution. Exact budgets and timing are selected during calibration and then frozen.
Balanced task conditions	Disaster/no disaster; correct/incorrect focal claims. Rotate source position and neutral identifiers using a prespecified schedule.
Claimed source	Optional extension: ordinary peer, relevant emergency expert, political authority, or religious authority. Initial pilot uses an ordinary peer.
Language	Optional matched English, Chinese, and Vietnamese conditions. Initial pilot uses English.
Model	Optional replications across documented models from different developers. Initial pilot uses one model/version; its selection remains open.
Study-specific treatment	The intervention table below defines the primary comparison for this proposal.

Source attribution. Change only a peer-visible claimed-source label while holding the focal message, evidence, actual permissions, delivery schedule, and position constant within matched comparisons. A title does not grant access to simulator truth. Distinguish a religious role (such as a monk or pastor), a named figure (such as the Buddha), and a divine-authority claim; do not pool these as equivalent. A named political figure such as Xi Jinping is likewise an optional identity condition, not the definition of political authority. Specific identities and wording must be selected before evaluation. Record refusals and role-play responses, since these may differ from accepting the claim. Attribution effects do not establish religious belief, political allegiance, or human cultural psychology.

Language and model replications. Use bilingual review to preserve instructions, evidence, and source meaning across translations; keep structured action labels constant. Hold one language throughout each episode initially. Compare homogeneous networks of each selected model before considering mixed-model networks. Record developer, exact model identifier/version, access date, and generation settings. Differences between developers' models cannot be attributed causally to national origin, training culture, or any single model property.

Staged evaluation. First calibrate task comprehension, then run the study-specific contrast with the ordinary-peer label, one model, and English. Next vary claimed source across correct and incorrect claims; subsequently replicate selected contrasts across languages and models. Do not run an unspecified full factorial crossing every factor. Prespecify hypotheses, contrasts, sample allocation, and multiplicity handling for each stage, separating confirmatory comparisons from exploratory ones. These optional factors are included in the research scope but are not completed experiments or a committed evaluation matrix.

3.3 Warning schedule and consultation access

Primary variable	Conditions
Focal-message repetition	Once / repeated, with the initial message held constant.
Reception policy	Fixed / agent-selected, with equal incoming message caps.
Accuracy controls	Correct warnings and reassurance; incorrect warnings and reassurance in separately balanced task conditions.

Repetition is the assigned treatment. Whether a message actually disagrees with a recipient is an observed property, not guaranteed by repetition. Log each recipient's pre-message Safe/Danger assessment and distinguish disagreement from repeated agreement. Analyze all assigned episodes primarily; any subgroup defined by an initial, pre-treatment assessment is secondary and prespecified. Do not condition causal claims on disagreement emerging after treatment.

Cross a once-only versus repeated warning schedule with fixed versus agent-selected reception. In the repeated condition the informant transmits the same observation on each eligible round; in the once-only condition it transmits that observation initially, then a neutral status signal. The schedule is controlled by the simulator. All other agents choose messages and actions live. This is a live network with one controlled source, not an entirely unconstrained society.

Deliver the initial warning to all four neighbours of its source in every condition. From the next round, each recipient receives messages from two of its four potential neighbours. Fixed reception uses a preselected balanced schedule; chosen reception lets recipients select their next-round sources before seeing those messages. Both conditions have the same incoming message cap. Log potential, selected, sent, and delivered links separately. Exclusion means not selecting an available source; evacuation-related silence is a distinct event. Include a matched random-selection diagnostic to check whether mere sparsity explains failures.

Cross claim accuracy with disaster state: correct Danger and incorrect Safe in disaster episodes, correct Safe and incorrect Danger in safe episodes. Balance these conditions and keep the source label and message format constant. The informant privately receives simulator truth, but its outward message is an ordinary peer claim without a truth certificate; the controlled schedule determines whether that claim matches the observation. This prevents authentication from confounding the accuracy comparison. Rotate identities and informant positions. Keep the source active during a short, prespecified warning window in every arm; this exception to voluntary evacuation is an explicit intervention. After that window, it follows the same action rules as the other agents.

3.4 Outcomes and inference

Primary outcomes are the fraction of eligible consultation opportunities selecting the source and the episode-level fraction of uninformed agents remaining at disaster. Report unnecessary evacuations in safe episodes and total payoff alongside them. Secondary measures include warning delivery, relaying, and time to evacuation. A message reference measures observable use, not internal attention.

Estimate the persistence contrast within each reception condition and their interaction, pairing state, graph, source position, and simulator seed across arms. Repeat independent model runs and report uncertainty at the episode level; nodes are not independent experimental units. The routing intervention tests whether access policy changes outcomes. Correlation between source rejection and failure does not prove mediation or dislike. Before increasing scale, check whether any access loss occurs, whether delivered warnings change actions, and whether both correct and incorrect warnings are treated sensibly.

4 Limitations and research status

This is an initial proposal, not a completed experiment or demonstrated research gap. The repository now includes an initial calibration simulator and Adaption runner; it does not yet implement every proposed reference condition. Offline fixtures and live smoke tests verify execution, not research hypotheses. Freeze model version, prompts, round distribution, sampling settings, sample size, invalid-output rules, and stopping criteria after separate calibration. Report invalid outputs and failed runs rather than silently dropping them. No hidden chain-of-thought is required. Use inert simulator actions and no external tools. Claimed-source, language, and model factors are specified as

staged extensions rather than simultaneous first-pilot manipulations. They test sensitivity to controlled inputs and model configurations, not cultural or religious traits.

The human task supplies an experimental foundation, not evidence about model behavior. Results would depend on rewards, context management, graph structure, and the changed timing protocol. A ring is a starting condition, not a general account of all networks. No claim of first-of-kind work is made.

5 Provenance

The user's friend's six-page *Authority Across Languages: Evaluating Socially Framed Prompt Injection and Monitoring in Multi-Agent AI Systems* informed the earlier authority-and-monitoring discussion. It is an unpublished proposal, not experimental evidence. The linked Google document is a submission template. This AI-assisted draft requires team review, real authorship information, and current submission-format checks before submission.

References

[1] H. Shirado, F. W. Crawford, and N. A. Christakis (2020). *Collective communication and behaviour in response to uncertain 'Danger' in network experiments.* Proceedings of the Royal Society A, 476:20190685. https://doi.org/10.1098/rspa.2019.0685.

[2] S. Schachter (1951). *Deviation, rejection, and communication.* Journal of Abnormal and Social Psychology, 46:190–207. https://doi.org/10.1037/h0062326. Original experimental report also appears in *Theory and Experiment in Social Communication* (1950), https://cdn.isr.umich.edu/pubFiles/historicPublications/Theory_4867_.PDF.

[3] A. N. Tak et al. (2026). *From Process Loss to Assembly Bonus: Human-Grounded Diagnosis of Multi-Agent LLM Collaboration.* https://arxiv.org/html/2609.13261v1.